American Computer Science League

2024 -2025 ● Contest 2: Short Problems Solutions ● Classroom Division

1. Prefix/Infix/Postfix Notation

3 + 4 * (8 − 4 / 2)
= 3 + 4 * (8 - (/42))
= 3 + 4 * (- 8 / 4 2)
= 3 + (* 4 - 8 / 4 2)
= + 3 * 4 - 8 / 4 2

B. + 3 * 4 − 8 / 4 2

2. Prefix/Infix/Postfix Notation

9 8 6 2 − * 2 / + 6 3 / −
= 9 8 (6 2 −) * 2 / + (6 3 /) −
= 9 (8 4 *) 2 / + 2 −
= 9 (32 2 /) + 2 −
= (9 16 +) 2 −
= 25 2 −
= 23

D. 23

3. Bit-String Flicking

01011 OR NOT 01101
= 01011 OR (NOT 01101)
= 01011 OR 10010
= 11011

A. 11011

4. Bit-String Flicking

(LSHIFT-2 01101) OR (RCIRC-3 10110) AND (RSHIFT-1 11011)
= 10100 OR (11010 AND 01101)
= 10100 OR 01000
= 11100

C. 11100

5. Prefix/Infix/Postfix Notation

+ − * 2 3 / 4 2 / * 8 + 1 3 4
= + − (* 2 3)(/ 4 2) / * 8 (+ 1 3) 4
= + (− 6 2) / (* 8 4) 4
= + 4 (/ 32 4)
= + 4 8
= 12

E. 12

6. Prefix/Infix/Postfix Notation

$$\frac{xy + z}{a^2}$$

= (x*y + z) / (a^2)
= (x y * z +) / (a^2)
= (x y * z +) / (a 2 ^)
= (x y * z +) (a 2 ^) /
= x y * z + a 2 ^ /

C. x y * z + a 2 ^ /

7. Bit-String Flicking

10110 OR 11011 AND 00101 XOR 01101
= 10110 OR ((11011 AND 00101) XOR 01101)
= 10110 OR (00011 XOR 01101)
= 10110 OR 01110
= 11110

A. 11110

8. Bit-String Flicking

Let X = abcde and NOT X = ABCDE
LHS = (NOT (RCIRC-3 (LSHIFT-2 X)))
= (NOT (RCIRC-3 (LSHIFT-2 abcde)))
= (NOT (RCIRC-3 cde00))
= NOT e00cd
= E11CD
RHS = 01100
LHS = RHS ⇒ E11CD = 01100 ⇒ a = *, b = *, C = 0, D = 0, E = 0
Therefore X = **111

D. **111

9. LISP (ADD 2 (SUB 7 3) 3 (MULT 2 5) 8) = (ADD 2 7÷3 3 2*5 8) = (ADD 2 4 3 10 8) = 2 + 4 + 3 + 10 + 8 = 27	E. 27
10. LISP (CDR (CAR (CDR (CDR '(e (f g) (d b (a c))))))) = (CDR (CAR (CDR '((f g) (d b (a c)))))) = (CDR (CAR '((d b (a c))))) = (CDR '(d b (a c))) = (b (a c))	B. (b (a c))